

Direct observation of meow meow in the presence of fart

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Abstract

The goal of this experiment is to measure X-ray diffraction patterns of various solid samples in order to describe their crystal structure at the atomic/molecular level. Any crystalline solid consists of a regular 3-dimensional packing of cells, the shape of which corresponds to one of fourteen Bravais lattices. These 3D patterns create multiple planar layers of various orientations within the crystal, each capable of producing a peak in the observed diffraction pattern. From that peak, Bragg's law $n\lambda = 2d \sin \theta$ (where $n = 1$ for a first-order plane) can be used to deduce the inter-planar distance d of each diffraction plane of the crystal. Using this, if the Miller indices of the different planes are known, one can identify the Bravais lattice and calculate the lattice parameter(s) of the crystal. The first samples to be observed are the pure copper and nickel ones, for which the obtained results (...). Following that, an alloy sample of Cu and Ni is observed, yielding (...). etc.

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1 Introduction

You can write a section intro here. Section 1.1 below shows a demo of a subsection.

This is a new paragraph, with some $M_{\text{ath}} = I_n \text{lin}_e$, and a larger equation can be made as follows:

$$ax^2 + bx + c = 0 \tag{1}$$

Sadly, Eq. 1 is super boring. Note the little % symbols in the latex code. They prevent a new paragraph from forming.

1.1 Test Subsection

You can also do subsubsections if you really want.

Figure ??(a) shows a simplified drawing of the Space Laser Facility (SPL). A bank of 10^{18} 9-volt batteries (Eveready ®X-treme bulkbox) supplies power to a sophisticated heat beam known as a “Laser”. In Fig. ??(b), we show the estimated response of earth for a laser swept along the equator at a rate of 5 radians / hour from east to west. Blue arrows indicate the approximate direction of motion. Due to the rotation of the earth, a similar sweep from west to east will not produce a clean cut [?]. The wavelength of this laser, $\lambda = 1.55 \mu\text{m}$, is chosen primarily because the telecom industry has produced a multitude of low-cost components at this wavelength.

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